

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		Fuel is burnt in the boiler or water is heated in the boiler (1) <u>Steam</u> {drives / turns} a turbine (1) Turbine { <u>drives</u> / <u>turns</u> } a generator (1) accept spins	3			3		
		(ii)		Electrical energy = 70 [J] (1) $\frac{70}{200} \times 100$ (1) = 35[%] (1) [so agree with Neve] Alternative: Energy loss = 130 [J] (1) $\frac{130}{200} \times 100$ (1) = 65[%] so 100 – 65 = 35[%] (1) [so agree with Neve] Award 2 marks for an answer of 65[%] or 0.35 Award 1 mark for an answer of 0.65			3	3	3	
	(b)	(i)		{Minimum / lowest / smallest} {output / power / supplied / demand / on the graph} accept energy	1			1		
		(ii)		40 [GW]		1		1	1	
		(iii)		Maximum output from Dinorwig = 1.8 GW (1) Peak demand is {4 h / from 16:00 to 20:00} (1) don't accept 2 different time periods N.B. 7.2 GWh seen award the 1 st two marks Extra demand [above 35 GW] is 5 GW (1) 3.2 GW is required from overseas (1) [so agree with Rowan] N.B. numbers have to be used in the correct context			4	4	4	
				Question 2 total	4	1	7	12	8	0